

# Project Assignment 0

CSE5825 BML

We will reinforce the probabilistic modelling concepts we learn in class through the final project which consists of four assignments. This first project assignment encourages you to start thinking about research problems in your area of expertise or interest that can be modelled probabilistically. Which scientific problems or hypotheses do you find interesting or care about? If your ideas on a project align with other students in the class, you are permitted to work in groups of 2-3. Group projects will be held to a higher standard than individual projects.

The only requirement for the selection of your project is that it builds from or uses ideas related to what we cover in class. For example, projects can be

1. *model development*. Reason about data (ideally in your area). What research questions can you formulate on this data? What are the variables? Which are observed or unobserved? How do they related to each other? How do you perform inference and model checking?
2. *algorithmic*. Develop new inference algorithms that can be applied to a large class of probabilistic models. Evaluate them theoretically or empirically.
3. *theoretical*. Further understanding the properties of a class of probabilistic models, or inference methods.
4. *review*. Perform an in-depth literature survey of a class of models or inference algorithms related to what we discussed in class.

Note that these categories are not mutually exclusive and, ideally, your project would incorporate elements from each.

**Your assignment is to identify data and describe three hypotheses that the data may support.**

The rest of your project will depend heavily on this assignment. You will be asked to **submit a short description of your data and hypotheses** to HuskyCT and **give a short  $\approx$  3-minute presentation (see course schedule)** to share your hypotheses and planned data set. This presentation will give an opportunity for your classmates to provide constructive feedback, identify areas of common interest, and form groups. If you know which group you want to work with, you can present as a group, but everyone must submit a written assignment completed on their own.

There are many ways to identify data and formulate hypotheses. You can get inspired by studies that deal with problems of interest. Use Google scholar or view papers from relevant conferences, including [NeurIPS](#) (check workshops too), [AISTATS](#), [ICML](#). You can search in both the current and previous years conferences and workshops. Alternatively, you could search through data sets from the resources below. Think about what types of hypotheses you can generate from the data.

# 1 Data resources

Public experimental data is preferred. Simulated data is sufficient although it is best used in conjunction with experimental data. Generative model formulations yield a method for simulation, but simulating from and evaluating on the same model might not be sufficient evidence to trust your model to a suspicious reviewer. Private or restricted data can also be used, but make sure access can be obtained in time to finish your project and all data usage requirements are met. A list of data for various scientific domains is included below.

## 1.1 General

1. [KDNuggets](#)
2. [Google Data Search](#)
3. [Google Datasets](#)
4. [Hadoop public data](#)
5. [Quora Where can I get data?](#)

## 1.2 AI and machine learning

1. [Hugging Face](#)
2. [Kaggle](#)
3. [UCI Machine Learning Repository](#)
4. [Scikit learn datasets](#)
5. [Imagenet](#)
6. [MNIST handwritten digits](#)
7. [Great article on data sources for ML.](#)

## 1.3 Examples of biological data

1. dbGaP: repository for genotype and other genomic data
2. Swiss-Prot/UniProt: database of gene transcripts, well annotated with functional information
3. Protein Data Bank (PDB): database of tertiary structures of tens of thousands of proteins

4. Gene Ontology: ontology represented as a directed acyclic graph of gene function, cellular location, biological role of many genes
5. Gene Expression Omnibus (GEO): gene expression and RNA-sequencing repository
6. The Cancer Genome Atlas (TCGA): gene expression and RNA-seq repository for cancer studies
7. Sage Bionetworks Synapse: repository for many different types of genomic study data
8. Ensembl: gene transcripts and isoforms, well annotated
9. 1000 Genomes/HapMap: 40M SNPs from > 1000 individuals worldwide, well curated and phased
10. NHGRI GWAS Catalog: all trait-associated SNPs ever published
11. GenBank (NCBI): database of all publicly available annotated genomic sequences
12. ENCODE data: repository for thousands of regulatory element assays on many different cell types
13. UCSC Genome Browser: visualization for all of these data sets and more

## **1.4 Miscellaneous Data**

1. Sociological data: [Fragile Families study](#).
2. Legal data: [Connecticut court and judicial data](#).
3. [Connecticut State Agencies Public Data](#).